

# Primary VPT Chain: ProducerUA -> ProZorro -> Retail

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely I(1)-like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

standardized\_type=other

retailer=silpo

promo\_variant=promo\_controlled

frequency=weekly

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo\_novus. Combined rule: daily median of available silpo and novus standardized\_type prices.

Table: PreTests

|                      | series               | adf_level_p | kpss_level_p | adf_diff1_p | kpss_diff1_p | adf_diff2_p | kpss_diff2_p | integration_class | stability_flag | standardized_type | retailer | promo_variant    | cointegration_p |
|----------------------|----------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------------|----------------|-------------------|----------|------------------|-----------------|
| producer_to_prozorro | producer             | nan         | nan          | nan         | nan          | nan         | nan          | ambiguous         | 0.0            | other             | silpo    | promo_controlled | nan             |
|                      | prozorro             | nan         | nan          | nan         | nan          | nan         | nan          | ambiguous         | 0.0            | other             | silpo    | promo_controlled | nan             |
|                      | retail               | nan         | nan          | nan         | nan          | nan         | nan          | ambiguous         | 0.0            | other             | silpo    | promo_controlled | nan             |
|                      | producer_to_prozorro | nan         | nan          | nan         | nan          | nan         | nan          | pair              | nan            | other             | silpo    | promo_controlled | nan             |
|                      | prozorro_to_retail   | nan         | nan          | nan         | nan          | nan         | nan          | pair              | nan            | other             | silpo    | promo_controlled | nan             |
|                      | producer_to_retail   | nan         | nan          | nan         | nan          | nan         | nan          | pair              | nan            | other             | silpo    | promo_controlled | nan             |

Table: ModelCoefficients

|                   |          |                  |           |      |              |          |          |       |         |         |          |            |                   |                  |           |              |                 |                              |                              |
|-------------------|----------|------------------|-----------|------|--------------|----------|----------|-------|---------|---------|----------|------------|-------------------|------------------|-----------|--------------|-----------------|------------------------------|------------------------------|
| standardized_type | retailer | promo_variant    | frequency | link | model_family | y_series | x_series | n_obs | sr_coef | lr_coef | ect_coef | ect_pvalue | asymmetry_short_p | asymmetry_long_p | vecm_rank | model_status | unreliable_flag | notes                        | primary_chain_doc            |
| other             | silpo    | promo_controlled | weekly    | none | none         | n/a      | n/a      | 0     | nan     | nan     | nan      | nan        | nan               | nan              | nan       | unavailable  | 1               | No admissible model estim... | Primary chain is strictly... |

Table: ResidualDiagnostics

|              |      |          |          |            |        |      |                 |
|--------------|------|----------|----------|------------|--------|------|-----------------|
| model_family | link | y_series | x_series | ljungbox_p | arch_p | jb_p | unreliable_flag |
| none         | none | n/a      | n/a      | nan        | nan    | nan  | 1               |

**Table: SeriesUsed**

No rows

# Table: LagProfile

No rows

Table: ModelEligibility

|                   |          |                  |           |                      |                      |                    |        |                              |
|-------------------|----------|------------------|-----------|----------------------|----------------------|--------------------|--------|------------------------------|
| standardized_type | retailer | promo_variant    | frequency | integration_producer | integration_prozorro | integration_retail | any_i2 | eligibility_note             |
| other             | silpo    | promo_controlled | weekly    | ambiguous            | ambiguous            | ambiguous          | 0      | Insufficient common sampl... |

**Table: NARDL\_Multipliers**

No rows



**Table: VECM\_IRF**

No rows

time\_series\_promo\_controlled.png

other | silpo | promo\_controlled

